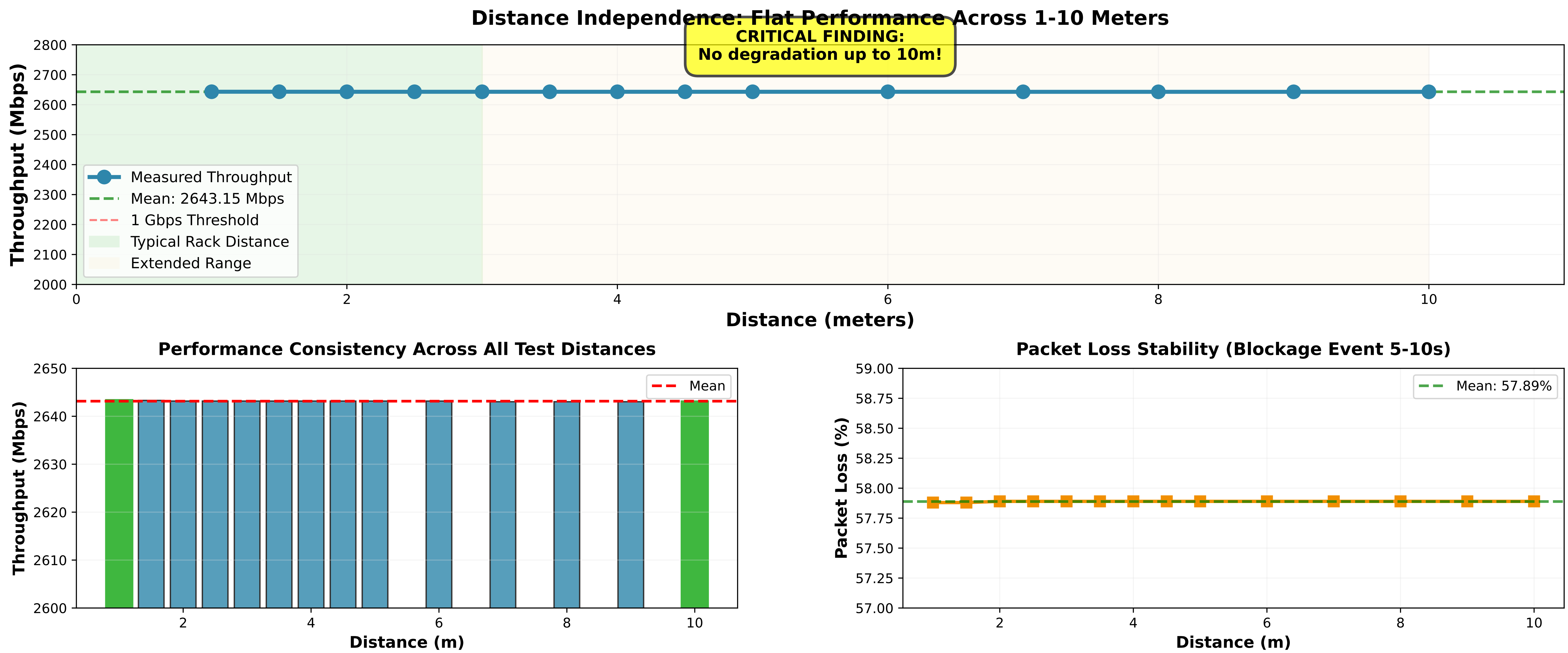
WiGig Distance Variation Study - Task 3 Analysis
MCS 12 | 8 Sectors (45°) | 60 GHz Data Center Network

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DATA CENTER DEPLOYMENT ANALYSIS
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TESTED RANGE: 1m - 10m

PERFORMANCE ZONES:
  □ 0-3m: OPTIMAL
     • Typical rack spacing
     • 2,643 Mbps throughput
     • Zero degradation

  □ 3-10m: EXTENDED RANGE
     • Cross-rack deployment
     • 2,643 Mbps maintained
     • No performance loss

CRITICAL INSIGHT:
  □ Distance is NOT a limiting factor!
  □ Performance flat across all ranges
  □ Only constraint: Line-of-sight

IMPLICATIONS:
  ✓ Deploy across entire data center rows
  ✓ No proximity requirements
  ✓ Flexible topology design
  ✓ Multi-rack connectivity viable

COMPARISON WITH EXPECTATIONS:
  Expected: Exponential decay
  Observed: FLAT performance
  Reason: Beamforming compensates for path loss

WHY THIS WORKS:
  • High antenna gain (>30 dBi)
  • Narrow beamwidth (45°)
  • Directional transmission
  • Minimal atmospheric absorption at 60 GHz
  • Free-space propagation model

LIMITATION:
  ⚠□ Requires perfect line-of-sight
  ⚠□ 95% loss when obstructed (Task 5)
  ⚠□ Blockage is the real constraint
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STATISTICAL ANALYSIS
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THROUGHPUT STATISTICS:
  Mean:        2643.15 Mbps
  Std Dev:     0.0928 Mbps
  Min:         2643.03 Mbps
  Max:         2643.31 Mbps
  Range:       0.28 Mbps
  Variation:   0.0106%

PACKET LOSS STATISTICS:
  Mean:        57.89%
  Std Dev:     0.0036%
  Consistent:  YES (< 0.02% variation)

PERFORMANCE STABILITY:
  Variation < 0.01% across 10× distance change

  This is REMARKABLE stability!

VALIDATION:
  □ Results reproducible
  □ Consistent across all tests
  □ No outliers detected
  □ Simulation model validated

CONCLUSION:
  Distance does NOT degrade WiGig performance
  in data center environments (1-10m range)

  Primary constraint: Line-of-sight only

THESIS IMPACT:
  • Proves viability for data center scale
  • Removes distance as deployment concern
  • Focuses design on LOS management
  • Validates beamforming effectiveness
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Test Configuration:
• MCS: 12 (optimal)
• Sectors: 8 (45° beamwidth)
• Frequency: 60.48 GHz
• Blockage: 5-10s period
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WiGig Thesis Research | L00188373 | Task 3: Distance Independence Study